

AIGUL SHARIP

Email: sharipaigul@gmail.com

Mobile: + 7-778-913-04-23

LinkedIn/GitHub: aigulsharip

PROFILE

Bioinformatician and Software Engineer with expertise in multi-omics analysis, biostatistical modeling, machine learning, and large-scale biomedical data integration. Author of 12 peer-reviewed publications (five as first author) spanning genomics and transcriptomics. Focused on applying computational approaches to uncover disease mechanisms and advance precision medicine. Passionate about driving translational medicine through the integration of data science, AI, and scalable software systems.

WORK EXPERIENCE

Software engineer/ Bioinformatician,

Mar 2025 – Present

EPAM Systems, a leading global IT consultancy with offices in 50+ countries

Client: AstraZeneca, Oct 2025 - Jan 2026

- Developed and maintained SMOCS and API Wall applications within the ADP platform
- Designed and implemented RESTful APIs to facilitate seamless interactions
- Collaborated with cross-functional teams to ensure development efforts aligned with business requirements and project goals
- Conducted unit testing and created thorough documentation to ensure code quality and maintainability

Client: *Invitae*, clinical genomics company specializing in genetic diagnostics, Mar-Sep 2025

- Contributed to the development of [Genosity Integrated Genomics Toolkit](#), a large-scale platform for managing and analyzing NGS data
- Designed and implemented bioinformatics workflows and pipelines supporting clinical genomics and precision medicine applications
- Collaborated with cross-functional teams to align computational solutions with bioinformatics workflows and clinical data processing needs
- Support backend development using Java (Spring Boot) and optimized infrastructure with AWS

Technical Stack: *Java, Spring Boot, Python, R, Nextflow, AWS, Docker, Gitlab, Unix/Linux*

Bioinformatician, part-time

Mar 2019 – Present

Lab. of Bioinformatics and System Biology, Nazarbayev University

Astana, Kazakhstan

- Developed and deployed machine learning models for Alzheimer's disease prediction using cognitive scores and biomedical features) and build an interactive Shiny web application [AD Prediction app](#) for rapid model-based risk assessment
- Performed large-scale WGS and RNA-seq analysis of Kazakhstani individuals, identifying disease-associated variants and pathways
- Conducted statistical analysis of risk factors of the severity of coronary atherosclerosis from patient data
- Carried out genome-wide association studies (GWAS) to identify genetic variants linked to complex diseases and traits
- Co-authored 10+ research publications, 7+ international internships and conferences (USA, Germany, Japan) \

Technical Stack: *R, Unix/Linux, Python, multi-omics data analysis, biostatistics, ML models*

Software Engineer

Mar - Sep 2024

BTS Digital

Astana, Kazakhstan

- Contributed to the development of the *Ismet platform* for Kazakhtelecom Business users.
- Developed *Ismet Docs* platform to enhance electronic document management.

Technical Stack: *Java, Spring, SQL, JavaScript, Postgres, GitLab, MinIO, Jasper, Agile*

Software Engineer

Jul 2022 – Jan 2024

Datanomix/Motio

Dallas, Texas, USA

- Participated in the enhancement of [Soterre](#), version control and deployment tool, designed for Qlik users.
- Executed the integration of external Qlik Sense platform functionalities for more user-friendly experience

Technical Stack: *Java, Spring, JavaScript, ExtJS, SQL, Postgres, Tomcat, Docker, GitLab, Agile*

Research Assistant

Jan 2017 - Mar 2019

Lab. of Dr. Jeanette Kunz, Nazarbayev University

Astana, Kazakhstan

- Contributed to the delivery of the research project “Phosphoinositide phosphate signaling networks: A novel therapeutic opportunity in breast cancer”
- Performed molecular biology experiments: design PCR primers, PCR, transfection, cell culturing, microscopy, image analysis, FACS, DNA isolation etc.

EDUCATION

Nazarbayev University (NU)

2017 -2019

- Master of Biological Sciences, **GPA 4.0/4.0**, with the Highest distinction
- Honors: Dean's list for Spring & Fall 2018; **Received Honor Diploma from the President of Kazakhstan**

Nazarbayev University

2013 -2017

- Bachelor of Biological Sciences, GPA 3.66/4.0
- Honors: Dean's list for Fall 2015 and Fall 2017 semesters

TECHNICAL SKILLS

Software Engineering	Java, JavaScript, R, Python, Nextflow, Unix/Linux, CSS, SQL, Postgres, React
ML & DS skills	supervised/unsupervised ML, feature selection, predictive modeling, statistics
Bioinformatics	multi-omics data analysis, WGS, RNA-seq, GWAS data analysis
Frameworks	Spring Framework, Java EE, Git, REST API, JUnit, Postman, Docker, Kubernetes, Microservices, OpenFeign, Eureka, Apache Kafka, ElasticSearch
Languages	Fluent in Kazakh, Russian and English, Intermediate German, Basic French

RESEARCH EXPERIENCE

Bioinformatician at National Laboratory Astana, NU

Sep 2018 - Present

Prof. Dr. Ulykbek Kairov, Lab. of Bioinformatics and System Biology, NU

Astana, Kazakhstan

- conducted whole-genome sequencing and variant analysis of Kazakh individuals, identifying genetic variants linked to metabolic and neurodegenerative diseases
- performed transcriptome profiling of Kazakhstani ESCC patients, discovering key dysregulated pathways and potential biomarkers for early diagnosis and precision therapies.
- analyzed statistical risk factors influencing the severity of coronary atherosclerosis from patient data

Laboratory research for Master Thesis “The role of bacteria and HLA-B27*05 allele on the pathogenesis of spondylarthritis”

Aug 2017 - May 2019

Prof. Jeannette Kunz and Prof. Christian Schoenbach, NU

Astana, Kazakhstan

- immunoinformatics analysis of potential Brucella epitopes, cell-based epitope binding assays and FACS analysis of gut microbiota

Research internship at German Rheumatism Research Center (DKFZ)

Jul - Aug 2018

Lab. of Dr. Kruglov, DKFZ

Berlin, Germany

- analysis of gut microbiota of AS patients and healthy controls by staining and sorting of flow cytometry DB Influx, stimulation experiments with human PBMC cells

Research internship at Kumamoto University

Nov-Dec 2017

Prof. Christian Schoenbach, IRCSM, Kumamoto University

Kumamoto, Japan

- analysis of gut microbiota of AS patients and healthy controls by staining and sorting of flow cytometry DB Influx, stimulation experiments with human PBMC cells

Laboratory research for Bachelor Thesis “Regulation of Focal Adhesion Assembly and Lamellopodia Dynamics with WNT Signalling Ligands”

Aug 2016 - May 2017

Prof. Adarichev and Prof. Jeannette Kunz, NU

Astana, Kazakhstan

- cell culturing and analysis rheumatoid arthritis cell lines with immunofluorescence techniques, optimize high-resolution time-lapse microscopy, plasmid isolation

Research internship at University of California San Francisco (USCF)

May - Aug 2016

Lab. of Prof. Rosemary Akhurst, Helen Diller Comprehensive Cancer Center, USCF

San Francisco, USA

- gene cloning by expression vectors, restriction enzyme digestions, DNA extraction from tissue and genotyping, primary cell culturing, siRNA transfection, Western blot

EXTRACURRICULUM ACHIEVEMENTS

- Awarded comprehensive fellowship to participate the Yessenov Data Lab program, summer 2021
- Travel award from School of Science and Technology, NU for internship at DRFZ, Germany, July-August 2018
- Scholarship for Research Internship from IRCMS, Kumamoto University, Japan, Nov-Dec 2017
- Aegean Conference at the 8th International Conference on Autoimmunity, Rhodes, Greece, Sep 2017
- Third prize in the Excellent Article Award in the 5th Guangzhou International Symposium., China, Dec 2018
- Winner of Yessenov Foundation Summer Research Fellowship for internship at UCSF, USA, Jan 2016
- Silver medal for NU Biology team at International iGEM competition, Boston, USA, Sep 2015

PROJECTS & COURSES

Software Engineer training

Nov 2020 – Aug 2021

nFactorial School

- Completed nFactorial Start - Introduction to Computer Science, 8-week intensive program in Java, with **Gold Certificate of Achievement**, accomplished 11 Java projects
- Improved problem-solving skills at 8-week intensive nFactorial Algorithms and Data Structures course

Java Backend Developer training

Oct 2021 – Mar 2022

BitLab Academy & OneLab

- Acquired a solid understanding of front-end development essentials using HTML, CSS, and JavaScript
- Explored the core principles of back-end development utilizing Java EE and Spring Frameworks
- Engineered an **Online Pharmacy** web application leveraging Spring Boot, Spring Data, and Spring Security

Data Scientist training

Jun–Aug 2021

Yessenov Data Lab, 8-week summer school for Data Scientist

- Applied data preprocessing, exploratory analysis, and visualization techniques using Python and R
- Implemented supervised machine learning models and evaluated performance
- Gained practical exposure to neural networks, model selection, and statistical modeling

Certifications from completed courses: <https://github.com/aigulsharip/certificates>

PUBLICATIONS

1. Serikzhan, A., Daniyarov, A., ... **Sharip, A.** et al. Genomic landscape of the Great Steppe: Genetic variants in healthy Kazakh individuals. *Scientific Data* 12, 1679 (2025)
2. **Sharip, A.**, & Kunz, J. (2025). Mechanosignaling via Integrins: Pivotal Players in Liver Fibrosis Progression and Therapy. *Cells* 14(4), 266
3. **Sharip, A.**, Rakhimova, S., Molkenov, A., Ashenova, A., Kozhamkulov, U., Akhmetollayev, I., ... & Kairov, U. (2024). Transcriptome profiling and analysis of patients with esophageal squamous cell carcinoma from Kazakhstan. *Frontiers in Genetics*, 15, 1249751.
4. Kairov, U.*, Molkenov, A.*, **Sharip, A.***, Rakhimova, S., Seidualy, M., Rhie, A., ... & Akilzhanova, A. (2022). Whole-genome sequencing and genomic variant analysis of Kazakh individuals. *Frontiers in Genetics*, 13, 902804. * Equal contribution
5. **Sharip, A.**, Mukhatayev, Z., Chunikhina, D., ... & Kunz, J. (2022). Microorganisms in the Pathogenesis and Management of Ankylosing Spondylitis. In *Role of Microorganisms in Pathogenesis and Management of Autoimmune Diseases: Volume I: Liver, Skin, Thyroid, Rheumatic & Myopathic Diseases* pp. 459-487.
6. Mukhatayev, Z., **Sharip, A.**, Nurgozhina, A., ... & Kunz, J. (2022). Microorganisms in the Pathogenesis and Management of Spondyloarthritis. *Role of Microorganisms in Pathogenesis and Management of Autoimmune Diseases: Volume I: Liver, Skin, Thyroid, Rheumatic & Myopathic Diseases*, 419-458
7. Seisenova, A., Daniyarov, A., Molkenov, A., **Sharip, A.**, Zinovyev, A., Kairov, U. Meta-Analysis of Esophageal Cancer Transcriptomes Using Independent Component Analysis. *Frontiers in Genetics* 2021, 12
8. Karabayev, D., Molkenov, A., Yerulanuly, K., Kabimoldayev, I., Daniyarov, A., **Sharip, A.**, Seisenova, A.; Zhumadilov, Z.; Kairov, U. re-Searcher: GUI-based bioinformatics tool for simplified genomics data mining of VCF files. *PeerJ* 2021, 9, e11333
9. Kairov, U., Molkenov, A., Rakhimova, S., Kozhamkulov, U., **Sharip, A.**, Karabayev, D., Daniyarov, A., H.Lee, J., D.Terwilliger, J., Akilzhanova, A., Zhumadilov, Z. Whole-genome sequencing data of Kazakh individuals. *BMC Research Notes* 2021, 14, 45
10. **Sharip, A.**, & Kunz, J. (2020). Understanding the pathogenesis of spondyloarthritis. *Biomolecules*, 14, 1461.
11. Sharip, A., Abdukhakimova, D., Wang, X., Kim, A., Kim, Y., **Sharip, A.**, ... & Xie, Y. (2017). Analysis of origin and protein-protein interaction maps suggests distinct oncogenic role of nuclear EGFR during cancer evolution. *Journal of Cancer*, 8(5), 903
12. Kim, Y., Kim, A., Sharip, A., **Sharip, A.**, Jiang, J., Yang, Q., & Xie, Y. (2017). Reverse the resistance to PARP inhibitors. *International journal of biological sciences*, 13(2), 198.